

Contents

Preliminary Lemmas	2
Lemma P1	2
Lemma P2	3
Lemma P3	4
Chapter 1 Classical Algebra	6
Exercise 1.2	6
Exercise 1.5	7
Chapter 2 The Fundamental Theorem of Algebra	8
Exercise 2.1	8
Exercise 2.6	8
Chapter 3 Factorisation of Polynomials	9
Exercise 3.5(a)	9
Exercise 3.5(b)	9
Exercise 3.5(c)	10
Exercise 3.7	10
Exercise 3.12(a)	10
Exercise 3.12(b)	10
Exercise 3.12(c)	10
Exercise 3.12(d)	10
Exercise 3.12(e)	11
Exercise 3.12(f)	11
Exercise 3.12(g)	11
Exercise 3.12(h)	11
Exercise 3.12(i)	11
Exercise 3.12(j)	11
Chapter 4 Field Extensions	11
Exercise 4.1	11
Exercise 4.2(a)	12
Exercise 4.2(b)	12
Exercise 4.2(c)	12
Exercise 4.2(d)	13
Exercise 4.2(e)	13
Exercise 4.2(f)	13
Exercise 4.2(g)	13
Exercise 4.5(a)	13
Exercise 4.5(b)	13
Exercise 4.5(c)	14
Exercise 4.8	14

Preliminary Lemmas

Lemma P1

The subset $S = \{p + q\sqrt{2} + ri + si\sqrt{2} : p, q, r, s \in \mathbb{Q}\}$ of $\mathbb{C}$ is a subfield.

Proof. Let $p_1 + q_1\sqrt{2} + r_1i + s_1i\sqrt{2}, p_2 + q_2\sqrt{2} + r_2i + s_2i\sqrt{2} \in S$, where $p_1, q_1, r_1, s_1, p_2, q_2, r_2, s_2 \in \mathbb{Q}$. Note that

$$(p_1 + q_1\sqrt{2} + r_1i + s_1i\sqrt{2}) + (p_2 + q_2\sqrt{2} + r_2i + s_2i\sqrt{2}) = (p_1 + p_2) + (q_1 + q_2)\sqrt{2} + (r_1 + r_2)i + (s_1 + s_2)i\sqrt{2} \in S.$$

Next, we can compute the product using the following Python code.

```
1 import sympy as sp
2
3 p1, q1, r1, s1, p2, q2, r2, s2 = sp.symbols('p1 q1 r1 s1 p2 q2 r2 s2', real=True)
4 sqrt2 = sp.sqrt(2)
5 i = sp.I
6
7 expression = (p1 + q1*sqrt2 + r1*i + s1*i*sqrt2) * (p2 + q2*sqrt2 + r2*i + s2*i*sqrt2)
8
9 print(sp.expand(expression))
```

We get

$$\begin{aligned} & (p_1 + q_1\sqrt{2} + r_1i + s_1i\sqrt{2})(p_2 + q_2\sqrt{2} + r_2i + s_2i\sqrt{2}) \\ &= p_1p_2 + p_1q_2\sqrt{2} + p_1r_2i + p_1s_2\sqrt{2}i + p_2q_1\sqrt{2} + p_2r_1i + p_2s_1\sqrt{2}i + 2q_1q_2 \\ & \quad + q_1r_2\sqrt{2}i + 2q_1s_2i + q_2r_1\sqrt{2}i + 2q_2s_1i - r_1r_2 - r_1s_2\sqrt{2} - r_2s_1\sqrt{2} - 2s_1s_2 \\ &= (p_1p_2 + 2q_1q_2 - r_1r_2 - 2s_1s_2) + (p_1q_2 + p_2q_1 - r_1s_2 - r_2s_1)\sqrt{2} \\ & \quad + (p_1r_2 + p_2r_1 + 2q_1s_2 + 2q_2s_1)i + (p_1s_2 + p_2s_1 + q_1r_2 + q_2r_1)\sqrt{2}i \\ & \in S. \end{aligned}$$

Also, we have

$$-(p_1 + q_1\sqrt{2} + r_1i + s_1i\sqrt{2}) = (-p_1) + (-q_1)\sqrt{2} + (-r_1)i + (-s_1)i\sqrt{2} \in S$$

and

$$1 = 1 + 0\sqrt{2} + 0i + 0i\sqrt{2} \in S.$$

By the subring test, S is a subring of $\mathbb{C}$. We next find an inverse for any non-zero element in S to show that it is a subfield. Let $p + q\sqrt{2} + ri + si\sqrt{2} \in S$, where $p, q, r, s \in \mathbb{Q}$. We first define

$$\alpha = (p + q\sqrt{2} + ri + si\sqrt{2})(p + q\sqrt{2} - ri - si\sqrt{2})(p - q\sqrt{2} + ri - si\sqrt{2})(p - q\sqrt{2} - ri + si\sqrt{2})$$

and compute the product using the following Python code.

```
1 import sympy as sp
2
3 p, q, r, s = sp.symbols('p q r s', real=True)
4 sqrt2 = sp.sqrt(2)
5 i = sp.I
6
```

```

7 alpha = (p + q*sqrt2 + r*i + s*i*sqrt2) * (p + q*sqrt2 - r*i - s*i*sqrt2) * (p -
  ↪ q*sqrt2 + r*i - s*i*sqrt2) * (p - q*sqrt2 - r*i + s*i*sqrt2)
8
9 print(sp.expand(alpha))

```

We get

$$\alpha = p^4 - 4p^2q^2 + 2p^2r^2 + 4p^2s^2 - 16pqrs + 4q^4 + 4q^2r^2 + 8q^2s^2 + r^4 - 4r^2s^2 + 4s^4 \in S.$$

Note that $p + q\sqrt{2} + ri + si\sqrt{2} = 0$ if and only if $p = q = r = s = 0$, if and only if $p + q\sqrt{2} - ri - si\sqrt{2} = p - q\sqrt{2} + ri - si\sqrt{2} = p - q\sqrt{2} - ri + si\sqrt{2} = 0$. Hence, $\alpha = 0$ if and only if $p + q\sqrt{2} + ri + si\sqrt{2} = 0$. Assume $p + q\sqrt{2} + ri + si\sqrt{2} \neq 0$. Dividing the product by α , we have

$$(p + q\sqrt{2} + ri + si\sqrt{2}) \left(\frac{(p + q\sqrt{2} - ri - si\sqrt{2})(p - q\sqrt{2} + ri - si\sqrt{2})(p - q\sqrt{2} - ri + si\sqrt{2})}{\alpha} \right) = 1.$$

So the inverse of $p + q\sqrt{2} + ri + si\sqrt{2}$ is $\frac{(p + q\sqrt{2} - ri - si\sqrt{2})(p - q\sqrt{2} + ri - si\sqrt{2})(p - q\sqrt{2} - ri + si\sqrt{2})}{\alpha}$. Since $p + q\sqrt{2} - ri - si\sqrt{2}, p - q\sqrt{2} + ri - si\sqrt{2}, p - q\sqrt{2} - ri + si\sqrt{2}, \alpha \in S$, the inverse is also an element of S , proving that S is a subfield. $\square$

Lemma P2

The subset $S = \{p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} : p, q, r, s \in \mathbb{Q}\}$ of $\mathbb{C}$ is a subfield.

Proof. Let $p_1 + q_1\sqrt{2} + r_1\sqrt{3} + s_1\sqrt{6}, p_2 + q_2\sqrt{2} + r_2\sqrt{3} + s_2\sqrt{6} \in S$, where $p_1, q_1, r_1, s_1, p_2, q_2, r_2, s_2 \in \mathbb{Q}$. Note that

$$\begin{aligned} & (p_1 + q_1\sqrt{2} + r_1\sqrt{3} + s_1\sqrt{6}) + (p_2 + q_2\sqrt{2} + r_2\sqrt{3} + s_2\sqrt{6}) \\ &= (p_1 + p_2) + (q_1 + q_2)\sqrt{2} + (r_1 + r_2)\sqrt{3} + (s_1 + s_2)\sqrt{6} \\ &\in S. \end{aligned}$$

Next, we can compute the product using the following Python code.

```

1 import sympy as sp
2
3 p1, q1, r1, s1, p2, q2, r2, s2 = sp.symbols('p1 q1 r1 s1 p2 q2 r2 s2')
4 sqrt2 = sp.sqrt(2)
5 sqrt3 = sp.sqrt(3)
6 sqrt6 = sp.sqrt(6)
7
8 product = (p1 + q1*sqrt2 + r1*sqrt3 + s1*sqrt6) * (p2 + q2*sqrt2 + r2*sqrt3 +
  ↪ s2*sqrt6)
9
10 print(sp.collect(sp.expand(product), [sqrt2, sqrt3, sqrt6]))

```

We get

$$\begin{aligned} & (p_1 + q_1\sqrt{2} + r_1\sqrt{3} + s_1\sqrt{6})(p_2 + q_2\sqrt{2} + r_2\sqrt{3} + s_2\sqrt{6}) \\ &= (p_1p_2 + 2q_1q_2 + 4r_1r_2 + 6s_1s_2) + (p_1q_2 + p_2q_1 + 3r_1s_2 + 3r_2s_1)\sqrt{2} \\ & \quad (p_1r_2 + p_2r_1 + 2q_1s_2 + 2q_2s_1)\sqrt{3} + (p_1s_2 + p_2s_1 + q_1r_2 + q_2r_1)\sqrt{6} \\ &\in S. \end{aligned}$$

Also, we have

$$-(p_1 + q_1\sqrt{2} + r_1\sqrt{3} + s_1\sqrt{6}) = (-p_1) + (-q_1)\sqrt{2} + (-r_1)\sqrt{3} + (-s_1)\sqrt{6} \in S$$

and

$$1 = 1 + 0\sqrt{2} + 0\sqrt{3} + 0\sqrt{6} \in S.$$

By the subring test, S is a subring of $\mathbb{C}$. We next find an inverse for any non-zero element in S to show that it is a subfield. Let $p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} \in S$, where $p, q, r, s \in \mathbb{Q}$. We first define

$$\alpha = (p+q\sqrt{2}+r\sqrt{3}+s\sqrt{6})(p+q\sqrt{2}-r\sqrt{3}-s\sqrt{6})(p-q\sqrt{2}+r\sqrt{3}-s\sqrt{6})(p-q\sqrt{2}-r\sqrt{3}+s\sqrt{6})$$

and compute the product using the following Python code.

```

1 import sympy as sp
2
3 p, q, r, s = sp.symbols('p q r s')
4 sqrt2 = sp.sqrt(2)
5 sqrt3 = sp.sqrt(3)
6 sqrt6 = sp.sqrt(6)
7
8 product = (p + q*sqrt2 + r*sqrt3 + s*sqrt6) * (p + q*sqrt2 - r*sqrt3 - s*sqrt6) * (p -
  ↪ q*sqrt2 + r*sqrt3 - s*sqrt6) * (p - q*sqrt2 - r*sqrt3 + s*sqrt6)
9
10 print(sp.expand(product))

```

We get

$$\alpha = p^4 - 4p^2q^2 - 6p^2r^2 - 12p^2s^2 + 48pqrs + 4q^4 - 12q^2r^2 - 24q^2s^2 + 9r^4 - 36r^2s^2 + 36s^4 \in S.$$

By Exercise 4.4, $1, \sqrt{2}, \sqrt{3}$ and $\sqrt{6}$ are linearly independent over $\mathbb{Q}$. Therefore, $p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} = 0$ if and only if $p = q = r = s = 0$, if and only if $p + q\sqrt{2} - r\sqrt{3} - s\sqrt{6} = p - q\sqrt{2} + r\sqrt{3} - s\sqrt{6} = p - q\sqrt{2} - r\sqrt{3} + s\sqrt{6} = 0$. Hence, $\alpha = 0$ if and only if $p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} = 0$. Assume $p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} \neq 0$. Dividing the product by α , we have $(p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6}) \left(\frac{(p+q\sqrt{2}-r\sqrt{3}-s\sqrt{6})(p-q\sqrt{2}+r\sqrt{3}-s\sqrt{6})(p-q\sqrt{2}-r\sqrt{3}+s\sqrt{6})}{\alpha} \right) = 1$.

So the inverse of $p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6}$ is $\frac{(p+q\sqrt{2}-r\sqrt{3}-s\sqrt{6})(p-q\sqrt{2}+r\sqrt{3}-s\sqrt{6})(p-q\sqrt{2}-r\sqrt{3}+s\sqrt{6})}{\alpha}$. Since $p + q\sqrt{2} - r\sqrt{3} - s\sqrt{6}, p - q\sqrt{2} + r\sqrt{3} - s\sqrt{6}, p - q\sqrt{2} - r\sqrt{3} + s\sqrt{6}, \alpha \in S$, the inverse is also an element of S , proving that S is a subfield. $\square$

Lemma P3

Let K be a subfield of $\mathbb{C}$. Then

$$\begin{aligned} K(\alpha) &= \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\} \\ &= \left\{ \frac{a_0 + \dots + a_n\alpha^n}{b_0 + \dots + b_m\alpha^m} : a_0, \dots, a_n, b_0, \dots, b_m \in K, b_0 + \dots + b_m\alpha^m \neq 0 \right\}. \end{aligned}$$

Proof. First, note that the second equality can be obtained by writing $p(\alpha) = a_0 + \dots + a_n\alpha^n$ and $q(\alpha) = b_0 + \dots + b_m\alpha^m$.

To prove the first equality, recall that $K(\alpha)$ is the smallest subfield of $\mathbb{C}$ generated by $K \cup \{\alpha\}$. Since $K \subseteq K(\alpha)$ and $\alpha \in K(\alpha)$, we have $k_0 + \dots + k_n \alpha^n \in K(\alpha)$ for any $k_0, \dots, k_n \in K$. If $k_0 + \dots + k_n \alpha^n \neq 0$, then $\frac{1}{k_0 + \dots + k_n \alpha^n} \in K(\alpha)$. Therefore, for any $a_0, \dots, a_n, b_0, \dots, b_m \in K$ with $b_0 + \dots + b_m \alpha^m \neq 0$, we have

$$\frac{a_0 + \dots + a_n \alpha^n}{b_0 + \dots + b_m \alpha^m} \in K(\alpha).$$

So

$$\left\{ \frac{a_0 + \dots + a_n \alpha^n}{b_0 + \dots + b_m \alpha^m} : a_0, \dots, a_n, b_0, \dots, b_m \in K, b_0 + \dots + b_m \alpha^m \neq 0 \right\} \subseteq K(\alpha).$$

Next, we want to prove that $\left\{ \frac{a_0 + \dots + a_n \alpha^n}{b_0 + \dots + b_m \alpha^m} : a_0, \dots, a_n, b_0, \dots, b_m \in K, b_0 + \dots + b_m \alpha^m \neq 0 \right\} = \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}$ is a subfield of $\mathbb{C}$. Let

$$\frac{p(\alpha)}{q(\alpha)}, \frac{\hat{p}(\alpha)}{\hat{q}(\alpha)} \in \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\},$$

where $q(\alpha), \hat{q}(\alpha) \neq 0$. Since K is a field and thus a domain, we have $q(\alpha)\hat{q}(\alpha) \neq 0$. Note that

$$\begin{aligned} \frac{p(\alpha)}{q(\alpha)} + \frac{\hat{p}(\alpha)}{\hat{q}(\alpha)} &= \frac{p(\alpha)\hat{q}(\alpha) + \hat{p}(\alpha)q(\alpha)}{q(\alpha)\hat{q}(\alpha)} \\ &\in \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}, \\ \frac{p(\alpha)}{q(\alpha)} \cdot \frac{\hat{p}(\alpha)}{\hat{q}(\alpha)} &= \frac{p(\alpha)\hat{p}(\alpha)}{q(\alpha)\hat{q}(\alpha)} \\ &\in \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}, \\ -\frac{p(\alpha)}{q(\alpha)} &= \frac{-p(\alpha)}{q(\alpha)} \\ &\in \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}. \end{aligned}$$

Also, since K is a subfield of $\mathbb{C}$, we have $1 \in K$. So $1 = \frac{1}{1} \in \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}$.

By the subring test, $\left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}$ is a subring of $\mathbb{C}$.

Now, suppose $\frac{p(\alpha)}{q(\alpha)} \neq 0$. If $p(\alpha) = 0$, then $\frac{p(\alpha)}{q(\alpha)} = \frac{0}{q(\alpha)} = 0$, a contradiction. Hence, $p(\alpha) \neq 0$. So we have $\frac{q(\alpha)}{p(\alpha)} \in \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}$. Note that $\frac{p(\alpha)}{q(\alpha)} \cdot \frac{q(\alpha)}{p(\alpha)} = 1$, so the inverse of $\frac{p(\alpha)}{q(\alpha)}$ is $\frac{q(\alpha)}{p(\alpha)}$. Therefore, $\left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}$ is a subfield of $\mathbb{C}$. Since $\left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}$ is a field containing $K \cup \{\alpha\}$, we have $K(\alpha) \subseteq \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}$. Hence,

$$K(\alpha) = \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\},$$

proving the first equality. □

Chapter 1 Classical Algebra

Exercise 1.2

Assume for contradiction that there exists $a, b \in \mathbb{Z}$ with $b \neq 0$ such that $\left(\frac{a}{b}\right)^2 = 2$. If a is negative and b is positive, then replacing a with $-a$ gives us $\left(\frac{-a}{b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $-a, b$ is also a valid pair of integers satisfying the conditions. If b is negative and a is positive, then replacing b with $-b$ gives us $\left(\frac{a}{-b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $a, -b$ is also a valid pair of integers satisfying the conditions. If both a and b are negative, then replacing a with $-a$ and b with $-b$ gives us $\left(\frac{-a}{-b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $-a, -b$ is also a valid pair of integers satisfying the conditions. Hence, we can assume without loss of generality that $a, b > 0$.

Since we assume that b only takes values in $\mathbb{N}$, by the well-ordering principle, there exists a smallest b satisfying the conditions. Note that

$$\begin{aligned}\left(\frac{2b-a}{a-b}\right)^2 &= \frac{4b^2 - 4ab + a^2}{a^2 - 2ab + b^2} \\&= \frac{4 - 4\left(\frac{a}{b}\right) + \left(\frac{a}{b}\right)^2}{\left(\frac{a}{b}\right)^2 - 2\left(\frac{a}{b}\right) + 1} \\&= \frac{4 - 4\left(\frac{a}{b}\right) + 2}{2 - 2\left(\frac{a}{b}\right) + 1} \\&= \frac{6 - 4\left(\frac{a}{b}\right)}{3 - 2\left(\frac{a}{b}\right)} \\&= \frac{6 - 4\left(\frac{a}{b}\right)}{3 - 2\left(\frac{a}{b}\right)} \times \frac{3 + 2\left(\frac{a}{b}\right)}{3 + 2\left(\frac{a}{b}\right)} \\&= \frac{18 + 12\left(\frac{a}{b}\right) - 12\left(\frac{a}{b}\right) - 8\left(\frac{a}{b}\right)^2}{9 - 4\left(\frac{a}{b}\right)^2} \\&= \frac{18 - 8 \times 2}{9 - 4 \times 2} \\&= 2.\end{aligned}$$

Recall that $\left(\frac{a}{b}\right)^2 = 2$, so we have $a^2 = 2b^2$ and thus $2b^2 - a^2 = 0$. Note that

$$\begin{aligned}(2b-a)(2b+a) &= 4b^2 - a^2 \\&= 2b^2 + (2b^2 - a^2) \\&= 2b^2 \\&> 0\end{aligned}$$

since $b > 0$ by assumption. Since $a, b > 0$, we have $2b + a > 0$. Hence, $2b - a > 0$. Also, from $a^2 = 2b^2$ we have $a^2 - b^2 = b^2$. Considering $b > 0$, we have

$$\begin{aligned}b^2 &> 0 \\a^2 - b^2 &> 0 \\(a-b)(a+b) &> 0 \\a-b &> 0\end{aligned}$$

since $a + b > 0$. From $2b - a > 0$ and $a - b > 0$ we know that $2b - a, a - b$ is also a pair of positive integers satisfying the conditions. However, from $2b - a > 0$ we have $a < 2b$, and hence $a - b < 2b - b = b$, contradicting the minimality of b . $\square$

Exercise 1.5

Let $S = \{p + q\alpha + r\alpha^2 \mid p, q, r \in \mathbb{Q}\}$. We first prove that S is a subring of $\mathbb{C}$.

Note that $1 = 1 + 0\alpha + 0\alpha^2 \in S$. Let $p + q\alpha + r\alpha^2, p' + q'\alpha + r'\alpha^2 \in S$, where $p, q, r, p', q', r' \in \mathbb{Q}$. Then we have $(p + q\alpha + r\alpha^2) + (p' + q'\alpha + r'\alpha^2) = (p + p') + (q + q')\alpha + (r + r')\alpha^2 \in S$, $-(p + q\alpha + r\alpha^2) = (-p) + (-q)\alpha + (-r)\alpha^2 \in S$ and $(p + q\alpha + r\alpha^2)(p' + q'\alpha + r'\alpha^2) = (pp' + 2rq') + (pq' + p'q + 2rr')\alpha + (pr' + qq' + qr' + rp')\alpha^2 \in S$. By the subring test, S is a ring.

Next, we construct an inverse explicitly to show that S is a subfield. Let $x = p + q\alpha + r\alpha^2 \in S \setminus \{0\}$, where $p, q, r \in \mathbb{Q}$. Then we have $x^2 = (p^2 + 4qr) + (2pq + 2r^2)\alpha + (2pr + q^2)\alpha^2$ and $x^3 = (p^3 + 12pqr + 2q^3 + 4r^3) + (3p^2q + 6pr^2 + 6q^2r)\alpha + (3p^2r + 3pq^2 + 6qr^2)\alpha^2$. Let $x^3 + k_2x^2 + k_1x + k_0 = 0$. Denote the coefficient of α^j in x^i by x_{ij} . Then we have $(x_{30} + x_{31}\alpha + x_{32}\alpha^2) + k_2(x_{20} + x_{21}\alpha + x_{22}\alpha^2) + k_1(p + q\alpha + r\alpha^2) + k_0 = 0$, that is, $(x_{30} + x_{20}k_2 + pk_1 + k_0) + (x_{31} + x_{21}k_2 + qk_1)\alpha + (x_{32} + x_{22}k_2 + rk_1)\alpha^2 = 0$. Since $1, \alpha, \alpha^2$ are linearly independent, comparing coefficients gives us the linear system

$$\begin{cases} k_0 + pk_1 + x_{20}k_2 = -x_{30} \\ qk_1 + x_{21}k_2 = -x_{31} \\ rk_1 + x_{22}k_2 = -x_{32} \end{cases}.$$

We can use the following python code to solve this system.

```

1  import sympy as sp
2
3  k0, k1, k2 = sp.symbols('k0 k1 k2')
4  p, q, r = sp.symbols('p q r')
5
6  x30 = p**3 + 12 * p * q * r + 2 * (q**3) + 4 * (r**3)
7  x31 = 3 * (p**2) * q + 6 * p * (r**2) + 6 * (q**2) * r
8  x32 = 3 * (p**2) * r + 3 * p * (q**2) + 6 * q * (r**2)
9  x20 = p**2 + 4 * q * r
10 x21 = 2 * p * q + 2 * (r**2)
11 x22 = 2 * p * r + q**2
12
13 M = sp.Matrix([
14     [1, p, x20, - x30],
15     [0, q, x21, - x31],
16     [0, r, x22, - x32]
17 ])
18
19 solution = sp.solve_linear_system(M, k0, k1, k2)
20
21 for var in [k0, k1, k2]:
22     print(f"{var} =")
23     sp.pprint(sp.simplify(solution[var]))

```

The solution is $k_0 = -p^3 + 6pqr - 2q^3 - 4r^3$, $k_1 = 3p^2 - 6qr$ and $k_2 = -3p$. Since $x^3 + k_2x^2 + k_1x + k_0 = 0$, we have $x(x^2 + k_2x + k_1 + k_0x^{-1}) = 0$. Recall that $\mathbb{C}$ is a field

and hence a domain, and $x \neq 0$. Therefore, $x^2 + k_2x + k_1 + k_0x^{-1} = 0$, that is,

$$x^{-1} = -\frac{1}{k_0}x^2 - \frac{k_2}{k_0}x - \frac{k_1}{k_0}.$$

Since x and x^2 are linear combinations of $1, \alpha, \alpha^2$, x^{-1} is also a linear combination of $1, \alpha, \alpha^2$. $\square$

Chapter 2 The Fundamental Theorem of Algebra

Exercise 2.1

We first prove the existence of such an expression. If $p(t)$ is constant, then set $p(t) = q(t)$ and this is the required form, since a constant polynomial $q(t)$ has no zeros in $\mathbb{Q}[t]$. If $p(t)$ is not constant, then $\partial p \geq 1$. By Proposition 2.7, there exist $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ and $0 \neq k \in \mathbb{C}$ such that $p(t) = k(t - \alpha_1) \dots (t - \alpha_n)$. The coefficient of t^n in the expression on the right hand side is k . Since $p(t) \in \mathbb{Q}[t]$, we have $k \in \mathbb{Q}$. Assume without loss of generality that $\alpha_1, \dots, \alpha_r \in \mathbb{Q}$ and $\alpha_{r+1}, \dots, \alpha_n \notin \mathbb{Q}$ for some $0 \leq r \leq n$. Let $q(t) = k(t - \alpha_{r+1}) \dots (t - \alpha_n)$. Then $p(t) = (t - \alpha_1) \dots (t - \alpha_r)q(t)$. Since $\alpha_{r+1}, \dots, \alpha_n$ are the only complex roots of $q(t)$, $q(t)$ has no roots in $\mathbb{Q}[t]$.

Suppose

$$p(t) = (t - \alpha_1) \dots (t - \alpha_r)q(t) = (t - \hat{\alpha}_1) \dots (t - \hat{\alpha}_s)\hat{q}(t)$$

such that $\alpha_1, \dots, \alpha_r, \hat{\alpha}_1, \dots, \hat{\alpha}_s \in \mathbb{Q}$ and $q(t), \hat{q}(t)$ have no zeros in $\mathbb{Q}[t]$. Assume without loss of generality that $\alpha_1 \leq \dots \leq \alpha_r$. Substitute $t = \alpha_1$, we get

$$0 = (\alpha_1 - \alpha_1) \dots (\alpha_1 - \alpha_r)q(\alpha_1) = (\alpha_1 - \hat{\alpha}_1) \dots (\alpha_1 - \hat{\alpha}_s)\hat{q}(\alpha_1).$$

Since $\hat{q}(\alpha_1) \neq 0$, we have $\alpha_1 - \hat{\alpha}_1 = 0$ or $\alpha_1 - \hat{\alpha}_2 = 0$ or $\dots$ or $\alpha_1 - \hat{\alpha}_s = 0$. Hence, $\alpha_1 = \hat{\alpha}_1$ or $\alpha_1 = \hat{\alpha}_2$ or $\dots$ or $\alpha_1 = \hat{\alpha}_s$. Similarly, by substituting $t = \alpha_2, \dots, \alpha_r$, we have $\alpha_1, \alpha_2, \dots, \alpha_r \in \{\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_s\}$. Assume without loss of generality that $\alpha_1 = \hat{\alpha}_1, \alpha_2 = \hat{\alpha}_2, \dots, \alpha_r = \hat{\alpha}_r$. Assume for contradiction that $s > r$, so $\hat{\alpha}_{r+1}, \dots, \hat{\alpha}_s$ exist. Since $(t - \alpha_1) \dots (t - \alpha_r) = (t - \hat{\alpha}_1) \dots (t - \hat{\alpha}_r) \neq 0$, we have $q(t) = (t - \hat{\alpha}_{r+1}) \dots (t - \hat{\alpha}_s)\hat{q}(t)$. Note that $\hat{\alpha}_{r+1}$ is a rational root of the polynomial on the right hand side, contradicting that $q(t)$ has no rational roots. Hence, $\hat{\alpha}_{r+1}, \dots, \hat{\alpha}_s$ do not exist and thus $q(t) = \hat{q}(t)$. This proves the uniqueness of such an expression.

Note that $p(\alpha_j) = (\alpha_j - \alpha_1) \dots (\alpha_j - \alpha_j) \dots (\alpha_j - \alpha_r)q(\alpha_j) = 0$, so α_j is a zero of $p(t)$ in $\mathbb{Q}$. Conversely, we want to prove that if $x \in \mathbb{Q}$ and $p(x) = 0$ then $x = \alpha_j$ for some j . Assume for contradiction that there exists $x \in \mathbb{Q}$ such that $p(x) = 0$ and $x \neq \alpha_j$ for all j . Then we have $x - \alpha_j \neq 0$ for all j . Since $q(t)$ has no zeros in $\mathbb{Q}[t]$, we have $q(x) \neq 0$. Since $\mathbb{Q}$ is a domain, we have $p(x) = (x - \alpha_1) \dots (x - \alpha_r)q(x) \neq 0$, which contradicts that x is a root of $p(t)$. Therefore, the α_j are precisely the zeros of $p(t)$ in $\mathbb{Q}$. $\square$

Exercise 2.6

Let $f(t)$ be a polynomial with $\partial f = n$. Suppose $f(t) = a_0 + a_1t + a_2t^2 + \dots + a_nt^n = 0$ for all $t \in \mathbb{C}$. Since $f(1) = 0$, we have

$$a_0 + a_1 + a_2 + \dots + a_n = 0.$$

Since $f(2) = 0$, we have

$$a_0 + 2a_1 + 2^2a_2 + \dots + 2^n a_n = 0.$$

Similarly, since $f(k) = 0$ for any $1 \leq k \leq n+1$, we have

$$a_0 + ka_1 + k^2a_2 + \dots + k^n a_n = 0.$$

Hence, we have the linear system

$$\begin{pmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2^2 & 2^3 & \cdots & 2^n \\ 1 & 3 & 3^2 & 3^3 & \cdots & 3^n \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n+1 & (n+1)^2 & (n+1)^3 & \cdots & (n+1)^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Note that

$$\begin{vmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2^2 & 2^3 & \cdots & 2^n \\ 1 & 3 & 3^2 & 3^3 & \cdots & 3^n \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n+1 & (n+1)^2 & (n+1)^3 & \cdots & (n+1)^n \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 3 & \cdots & n+1 \\ 1^2 & 2^2 & 3^2 & \cdots & (n+1)^2 \\ 1^3 & 2^3 & 3^3 & \cdots & (n+1)^3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1^n & 2^n & 3^n & \cdots & (n+1)^n \end{vmatrix},$$

which is the Vandermonde determinant with $z_1 = 1, z_2 = 2, \dots, z_{n+1} = n+1$. By Exercise 2.5, this determinant is non-zero. So the linear system has no non-trivial solutions. Hence, $a_0 = a_1 = a_2 = \dots = a_n = 0$. Therefore, all the coefficients of f are zero. $\square$

Chapter 3 Factorisation of Polynomials

Exercise 3.5(a)

Let $f(t) = t^4 + t^3 + t^2 + t + 1$. Note that $f(t)$ is irreducible over $\mathbb{Q}$ if and only if $f(t+1)$ is irreducible over $\mathbb{Q}$. Since

$$\begin{aligned} f(t+1) &= (t+1)^4 + (t+1)^3 + (t+1)^2 + (t+1) + 1 \\ &= t^4 + 5t^3 + 10t^2 + 10t + 5, \end{aligned}$$

by Eisenstein's criterion with $q = 5$, $f(t+1)$ is irreducible over $\mathbb{Q}$. Therefore, $t^4 + t^3 + t^2 + t + 1$ is also irreducible over $\mathbb{Q}$. $\square$

Exercise 3.5(b)

Let $f(t) = t^5 + t^4 + t^3 + t^2 + t + 1$. Note that $f(-1) = (-1)^5 + (-1)^4 + (-1)^3 + (-1)^2 + (-1) + 1 = 0$. By Remainder Theorem, $t+1$ is a factor of $f(t)$, so $f(t) = (t+1)g(t)$ for some $g(t) \in \mathbb{Q}[t]$. Therefore, $t^5 + t^4 + t^3 + t^2 + t + 1$ is reducible over $\mathbb{Q}$. $\square$

Exercise 3.5(c)

Let $f(t) = t^6 + t^5 + t^4 + t^3 + t^2 + t + 1$. Note that $f(t)$ is irreducible over $\mathbb{Q}$ if and only if $f(t+1)$ is irreducible over $\mathbb{Q}$. Since

$$\begin{aligned} f(t+1) &= (t+1)^6 + (t+1)^5 + (t+1)^4 + (t+1)^3 + (t+1)^2 + (t+1) + 1 \\ &= t^6 + 7t^5 + 21t^4 + 35t^3 + 35t^2 + 21t + 7, \end{aligned}$$

by Eisenstein's criterion with $q = 7$, $f(t+1)$ is irreducible over $\mathbb{Q}$. Therefore, $t^6 + t^5 + t^4 + t^3 + t^2 + t + 1$ is also irreducible over $\mathbb{Q}$. $\square$

Exercise 3.7

Suppose $f \neq \mathbf{0}$ is prime. Assume for contradiction that f is reducible over K . Let $f = gh$, where $g, h \in K[t]$ and $\partial g, \partial h < \partial f$. Since $f \mid f$, we have $f \mid gh$, so $f \mid g$ or $f \mid h$. Since $gh = f \neq \mathbf{0}$, by Exercise 3.12(e), we have $\partial f \leq \partial g$ or $\partial f \leq \partial h$, a contradiction.

Now, suppose $f \neq \mathbf{0}$ is irreducible. Define

$$\begin{aligned} I &= \{g \in K[t] : f \mid g\} \\ &= \{g \in K[t] : g = fq \text{ for some } q \in K[t]\} \\ &= \langle f \rangle. \end{aligned}$$

By Lemma RF 6.2, I is a maximal ideal of $K[t]$. If f is a constant, then f is prime because a non-zero constant divides any polynomial. Assume f is not constant. Then f does not divide 1, so I is a proper ideal. Since $K[t]$ is commutative, by Corollary RF 4.13, $K[t]/I$ is a field. Hence, $K[t]/I$ is an integral domain. By Theorem RF 4.16, I is a prime ideal. Therefore, for any $g, h \in K[t]$, $gh \in I$ implies $g \in I$ or $h \in I$, which means that $f \mid gh$ implies $f \mid g$ or $f \mid h$. Hence, f is prime. $\square$

Exercise 3.12(a)

The statement is false. A counter-example is $f(t) = t^2$, which is of degree 2 but has only one distinct zero, namely $t = 0$. $\square$

Exercise 3.12(b)

The statement is true. This is the same as Theorem 3.30, where the subfield is just $\mathbb{C}$ itself. $\square$

Exercise 3.12(c)

The statement is false. The counter-example from Exercise 3.12(a) also applies here. $\square$

Exercise 3.12(d)

The statement is false. Any non-zero polynomial f and $g = f$ is a counter-example. Clearly, $f \mid g$ and $\partial f \not\leq \partial g$. $\square$

Exercise 3.12(e)

The statement is true. If f, g are non-zero polynomials and $f \mid g$, then $g = qf$ for some $q \in \mathbb{C}[t]$. If $q = 0$, then $g = 0f = 0$, a contradiction. So $q \neq 0$ and thus $\partial q \geq 0$. Therefore, $\partial g = \partial q + \partial f \geq \partial f$. $\square$

Exercise 3.12(f)

The statement is true. Assume for contradiction that there exists a reducible polynomial f of degree 1. Let $f = gh$, where $g, h \in \mathbb{C}[t]$ and $\partial g, \partial h < \partial f = 1$. Hence, g and h are constant (zero or non-zero). Therefore, gh must be a constant, and thus has degree $-\infty$ or 0. So $\partial f = \partial(gh) \neq 1$, a contradiction. $\square$

Exercise 3.12(g)

Exercise 3.12(h)

The statement is true. This is the statement of Gauss's Lemma. $\square$

Exercise 3.12(i)

The statement is false. A counter-example is $f(t) = t^2 - 2$. Assume for contradiction that f is reducible over $\mathbb{Z}$. Then $t^2 - 2 = (at + b)(ct + d)$ for some $a, b, c, d \in \mathbb{Z}$. We can assume without loss of generality that $a = c = 1$, so $t^2 - 2 = (t + b)(t + d) = t^2 + (b + d)t + bd$. By comparing coefficients, we have $b + d = 0$ and $bd = -2$. Therefore, $b = -d$ and thus $-d^2 = -2$, which implies $d^2 = 2$. However, no integer d satisfies $d^2 = 2$. Hence, this is a contradiction. So f is irreducible over $\mathbb{Z}$. Since $t^2 - 2 = (t + \sqrt{2})(t - \sqrt{2})$, f is reducible over $\mathbb{R}$. $\square$

Exercise 3.12(j)

The statement is true. The contrapositive of the statement is "if a polynomial f with integer coefficients is reducible over $\mathbb{Z}$, then it is reducible over $\mathbb{R}$ ". Let f be a polynomial with integer coefficients that is reducible over $\mathbb{Z}$. Then $f = gh$ for some $g, h \in \mathbb{Z}[t]$ with $\partial g, \partial h < \partial f$. Since $g, h \in \mathbb{R}[t]$, f is reducible over $\mathbb{R}$. $\square$

Chapter 4 Field Extensions

Exercise 4.1

For field extensions ι and j , define $\iota \sim j$ as ι is isomorphic to j . Let $\iota_1 : K_1 \rightarrow \hat{K}_1$, $\iota_2 : K_2 \rightarrow \hat{K}_2$ and $\iota_3 : K_3 \rightarrow \hat{K}_3$ be field extensions.

For the reflexive property, we want to prove that there exists field isomorphisms $\lambda : K_1 \rightarrow K_1$ and $\mu : \hat{K}_1 \rightarrow \hat{K}_1$, such that

$$\iota_1(\lambda(k)) = \mu(\iota_1(k))$$

for all $k \in K_1$. Consider λ and μ defined by $\lambda(k) = k$ and $\mu(\hat{k}) = \hat{k}$ for all $k \in K_1$ and $\hat{k} \in \hat{K}$. Clearly, λ and μ are field isomorphisms. Also, for any $k \in K_1$, we have

$$\iota_1(\lambda(k)) = \iota_1(k) = \mu(\iota_1(k)).$$

Therefore, $\iota_1 \sim \iota_1$.

For the symmetric property, assume $\iota_1 \sim \iota_2$, so there exist field isomorphisms $\lambda_1 : K_1 \rightarrow K_2$ and $\mu : \hat{K}_1 \rightarrow \hat{K}_2$ such that $\iota_2(\lambda_1(k)) = \mu_1(\iota_1(k))$ for all $k \in K_1$. We want to prove that $\iota_2 \sim \iota_1$, that is, there exist field isomorphisms $\lambda_2 : K_2 \rightarrow K_1$ and $\mu_2 : \hat{K}_2 \rightarrow \hat{K}_1$ such that $\iota_1(\lambda_2(k)) = \mu_2(\iota_2(k))$ for all $k \in K_2$. Since λ_1 and μ_1 are bijective, they have inverses λ_1^{-1} and μ_1^{-1} . Let $\lambda_2 = \lambda_1^{-1}$ and $\mu_2 = \mu_1^{-1}$. Then

$$\begin{aligned}\iota_1(\lambda_2(k)) &= \iota_1(\lambda_1^{-1}(k)) \\ &= \mu_1^{-1}(\mu_1(\iota_1(\lambda_1^{-1}(k)))) \\ &= \mu_2(\iota_2(\lambda_1(\lambda_1^{-1}(k)))) \\ &= \mu_2(\iota_2(k))\end{aligned}$$

for any $k \in K_2$. Therefore, $\iota_2 \sim \iota_1$.

For the transitive property, suppose $\iota_1 \sim \iota_2$ and $\iota_2 \sim \iota_3$, so there exist field isomorphisms $\lambda_1 : K_1 \rightarrow K_2$, $\mu_1 : \hat{K}_1 \rightarrow \hat{K}_2$, $\lambda_2 : K_2 \rightarrow K_3$ and $\mu_2 : \hat{K}_2 \rightarrow \hat{K}_3$ such that $\iota_2(\lambda_1(k)) = \mu_1(\iota_1(k))$ for all $k \in K_1$ and $\iota_3(\lambda_2(k)) = \mu_2(\iota_2(k))$ for all $k \in K_2$. We want to prove that $\iota_1 \sim \iota_3$, that is, there exist field isomorphisms $\lambda_3 : K_1 \rightarrow K_3$ and $\mu_3 : \hat{K}_1 \rightarrow \hat{K}_3$ such that $\iota_3(\lambda_3(k)) = \mu_3(\iota_1(k))$ for all $k \in K_1$. Consider $\lambda_3 = \lambda_2 \circ \lambda_1$ and $\mu_3 = \mu_2 \circ \mu_1$. Note that λ_3 and μ_3 are field isomorphisms because the composition of field isomorphisms is also a field isomorphism. We have

$$\begin{aligned}\iota_3(\lambda_3(k)) &= \iota_3(\lambda_2(\lambda_1(k))) \\ &= \mu_2(\iota_2(\lambda_1(k))) \\ &= \mu_2(\mu_1(\iota_1(k))) \\ &= \mu_3(\iota_1(k))\end{aligned}$$

for any $k \in K_1$. Therefore, $\iota_1 \sim \iota_3$.

Hence, the isomorphism of field extensions is an equivalence relation. $\square$

Exercise 4.2(a)

Since $\{0, 1\} \subseteq \mathbb{Q}$ and $\mathbb{Q}$ is the smallest subfield of $\mathbb{C}$, the subfield of $\mathbb{C}$ generated by $\{0, 1\}$ is $\mathbb{Q}$. $\square$

Exercise 4.2(b)

Since $\{0\} \subseteq \mathbb{Q}$ and $\mathbb{Q}$ is the smallest subfield of $\mathbb{C}$, the subfield of $\mathbb{C}$ generated by $\{0\}$ is $\mathbb{Q}$. $\square$

Exercise 4.2(c)

Let L be the subfield generated by $\{0, 1, i\}$. Since $\mathbb{Q} \subseteq L$ and $i \in L$, we have $p + qi \in L$ for any $p, q \in \mathbb{Q}$, so $\{p + qi : p, q \in \mathbb{Q}\} \subseteq L$. Since $\{0, 1, i\} \subseteq \{p + qi : p, q \in \mathbb{Q}\}$ and $\{p + qi : p, q \in \mathbb{Q}\}$ is a field, $L \subseteq \{p + qi : p, q \in \mathbb{Q}\}$. Hence, the subfield of $\mathbb{C}$ generated by $\{0, 1, i\}$ is $\{p + qi : p, q \in \mathbb{Q}\}$. $\square$

Exercise 4.2(d)

Let L be the subfield generated by $\{i, \sqrt{2}\}$. Since $\mathbb{Q} \subseteq L$ and $i, \sqrt{2} \in L$, we have $i\sqrt{2} \in L$ and thus $p + q\sqrt{2} + ri + si\sqrt{2} \in L$ for any $p, q, r, s \in \mathbb{Q}$. Hence, $\{p + q\sqrt{2} + ri + si\sqrt{2} : p, q, r, s \in \mathbb{Q}\} \subseteq L$. Since $\{i, \sqrt{2}\} \subseteq \{p + q\sqrt{2} + ri + si\sqrt{2} : p, q, r, s \in \mathbb{Q}\}$ and $\{p + q\sqrt{2} + ri + si\sqrt{2} : p, q, r, s \in \mathbb{Q}\}$ is a field by Lemma P1, $L \subseteq \{p + q\sqrt{2} + ri + si\sqrt{2} : p, q, r, s \in \mathbb{Q}\}$. Hence, the subfield of $\mathbb{C}$ generated by $\{i, \sqrt{2}\}$ is $\{p + q\sqrt{2} + ri + si\sqrt{2} : p, q, r, s \in \mathbb{Q}\}$. $\square$

Exercise 4.2(e)

Let L be the subfield generated by $\{\sqrt{2}, \sqrt{3}\}$. Since $\mathbb{Q} \subseteq L$ and $\sqrt{2}, \sqrt{3} \in L$, we have $\sqrt{6} = \sqrt{2}\sqrt{3} \in L$ and thus $p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} \in L$ for any $p, q, r, s \in \mathbb{Q}$. Hence, $\{p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} : p, q, r, s \in \mathbb{Q}\} \subseteq L$. Since $\{\sqrt{2}, \sqrt{3}\} \subseteq \{p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} : p, q, r, s \in \mathbb{Q}\}$ and $\{p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} : p, q, r, s \in \mathbb{Q}\}$ is a field by Lemma P2, $L \subseteq \{p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} : p, q, r, s \in \mathbb{Q}\}$. Hence, the subfield of $\mathbb{C}$ generated by $\{\sqrt{2}, \sqrt{3}\}$ is $\{p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6} : p, q, r, s \in \mathbb{Q}\}$. $\square$

Exercise 4.2(f)

Since $\mathbb{R}$ is a field, the subfield of $\mathbb{C}$ generated by $\mathbb{R}$ is $\mathbb{R}$ itself. $\square$

Exercise 4.2(g)

Let L be the subfield generated by $\mathbb{R} \cup \{i\}$. Since $\mathbb{R} \subseteq L$ and $i \in L$, we have $a + bi \in L$ for any $a, b \in \mathbb{R}$. Hence, $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\} \subseteq L$. Since $L \subseteq \mathbb{C}$, the subfield of $\mathbb{C}$ generated by $\mathbb{R} \cup \{i\}$ is $\mathbb{C}$. $\square$

Exercise 4.5(a)

Since $\mathbb{Q}$ is defined as the equivalence classes of $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, we have $|\mathbb{Q}| \leq |\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})| = |\mathbb{Z}| \times |\mathbb{Z} \setminus \{0\}| = \aleph_0 \times \aleph_0 = \aleph_0$. Therefore, $\mathbb{Q}$ is countable. $\square$

Exercise 4.5(b)

Let K be a countable field and L/K be a simple extension. Then $L = K(\alpha)$ for some α . By Lemma P3, we have $K(\alpha) = \left\{ \frac{p(\alpha)}{q(\alpha)} : p(\alpha), q(\alpha) \in K[\alpha], q(\alpha) \neq 0 \right\}$, so $|K(\alpha)| \leq |K[\alpha]| \times |K[\alpha]|$. Since $K[\alpha] = \{k_0 + \dots + k_n\alpha^n : n \in \mathbb{N}, k_0, \dots, k_n \in K\}$, we have $|K[\alpha]| = |\mathbb{N}| \times |K| = \aleph_0 \times \aleph_0 = \aleph_0$. Therefore, $|K(\alpha)| \leq \aleph_0 \times \aleph_0 = \aleph_0$. Hence, $K(\alpha)$ is countable. $\square$

Exercise 4.5(c)

We first prove that $[0, 1)$ is uncountable. Assume for contradiction that it is countable. Then we can list all the elements in $[0, 1)$ indexed by $\mathbb{N}$. For example,

$$\begin{aligned}a_1 &= 0.1729384721 \dots \\a_2 &= 0.9983749283 \dots \\a_3 &= 0.1274812733 \dots \\a_4 &= 0.5723948374 \dots \\a_5 &= 0.6812036124 \dots \\&\vdots\end{aligned}$$

Denote the j -th digit after the decimal point of a_i by a_{ij} . We can construct a number $a \in [0, 1)$ such that the i -th digit of a is

$$\begin{cases} a_{ii} + 1 & \text{if } a_{ii} \neq 9 \\ 0 & \text{if } a_{ii} = 9 \end{cases}.$$

In this example, we have $a = 0.20841 \dots$. For any $i \in \mathbb{N}$, a is different from a_i by at least one digit, namely the i -th digit. Hence, a is not on the list, which contradicts that the list contains all elements in $[0, 1)$. Therefore, $[0, 1)$ is uncountable.

Since $\mathbb{R} \supseteq [0, 1)$, we have $|\mathbb{R}| \geq |[0, 1)| > \aleph_0$. Hence, $\mathbb{R}$ is uncountable. $\square$

Exercise 4.8

Assume for contradiction that t has a multiplicative inverse in $\mathbb{C}[t]$. Let $p(t) = a_0 + a_1t + a_2t^2 + \dots + a_nt^n \in \mathbb{C}[t]$ be the multiplicative inverse of t , where $a_0, a_1, a_2, \dots, a_n \in \mathbb{C}$. Then we have $tp(t) = 1$, so

$$a_0t + a_1t^2 + a_2t^3 + \dots + a_nt^{n+1} = 1.$$

The constant term of the expression on the left is 0, but that on the right is 1. Hence, this is a contradiction. So t has no multiplicative inverse in $\mathbb{C}[t]$. $\square$